Robust PAC Learning of Concurrent Stochastic Games

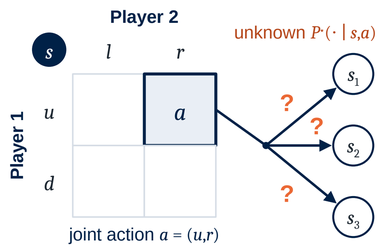
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NEURAL INFORMATION PROCESSING SYSTEMS

TL;DR **The first PAC-learning framework for two-player general-sum concurrent stochastic games (CSGs) with uncertain transitions**, returning, with probability ≥ 1 − δ, either an ε-Nash equilibrium (ε-optimal if an exact NE exists) or a **sound certificate** of stationary NE non-existence.

$\tilde{\mathcal{O}}\left(R_{\max}^2 H^4 |S|^2 |A| \,/\, (\varepsilon^2 p_{\text{reach}})\right)$ trajectory samples (Theorem 7)

Learning equilibria from samples



- **2-player CSG:** players act *simultaneously*; their joint action draws the next state from the true kernel $P^\star$.
- $P^\star$ **is unknown**.
- **Objectives:** probabilistic and reward reachability, given bounded rewards $|r_i| \le R_{\max}$.

How many interactions suffice to compute an approximately optimal equilibrium with high probability — and what happens when no exact equilibrium exists?

Two challenges

1 Equilibrium existence under uncertainty
Exact stationary NE may not exist [1]; even if one does, an uncertainty set around $P^\star$ may admit no *robust* NE [2].

2 Global coverage from stochastic trajectories
Sufficient joint state–action coverage must come from online trajectories, under joint control and uncertain transitions.

PAC-CSG: an equilibrium or a certificate

Nash margin — how far a stationary profile σ is from being an NE:

$$\mu(\sigma, P) \coloneqq \min_{i \in N} \min_{s \in S} \inf_{\sigma_i' \in \Sigma_i} [u_i(\sigma, P \mid s) - u_i(\sigma_{-i}[\sigma_i'], P \mid s)]$$

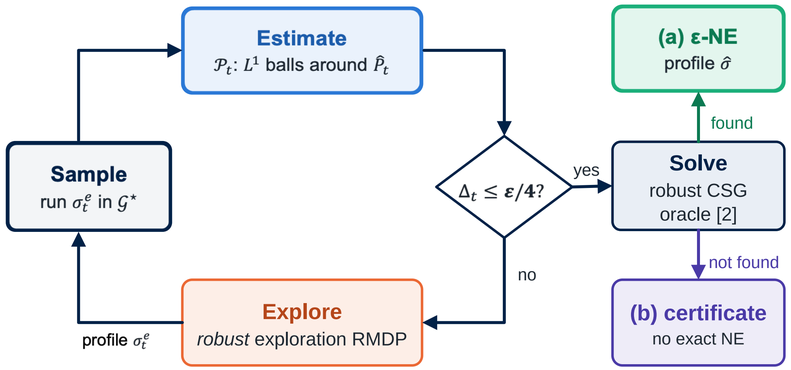
$$\mu(\sigma, P) \ge 0 \Leftrightarrow \sigma \text{ is an NE}$$
$$\bar{\mu} \coloneqq \max_\sigma \mu(\sigma, P^\star) \ge 0 \Leftrightarrow \text{true game } \mathcal{G}^\star \text{ has an exact NE}$$

PAC guarantee (Def. 4) — with probability ≥ 1 − δ, after polynomially many interactions, the output is:

(a) if an exact stationary NE exists: **an ε-NE** $\hat{\sigma}$ that is also **ε-optimal**, i.e.,
$$V^\star - u(\hat{\sigma}, P^\star) \le \varepsilon$$

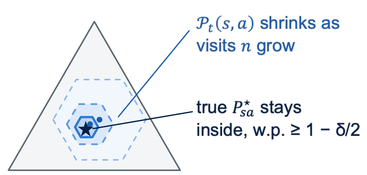
(b) a sound certificate that $\mathcal{G}^\star$ has **no exact stationary NE**

Algorithm: estimate, test, explore



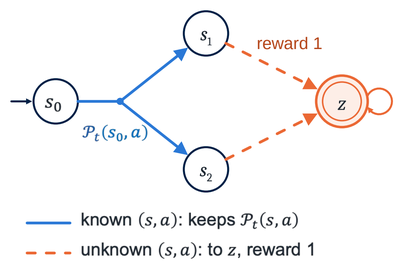
Exploration

L^1 confidence sets



$$\alpha_t(s,a) = \sqrt{\frac{2}{n} \ln \frac{2(2^{|S|} - 2)\,|S||A|\, n(n+1)}{\delta}}$$

Exploration RMDP



Why "robust" exploration? Conditioning on $P^\star \in \mathcal{P}_t$, the worst-case probability of reaching an unknown slot lower-bounds the true one:

$$p_t^e(P^\star) \;\ge\; \inf_{P \in \mathcal{P}_t} p_t^e(P) \;\ge\; p_{\text{reach}} \;>\; 0$$

Why it works

1 Sensitivity Lem. 2	$\lvert u_i(\sigma, P \mid s) - u_i(\sigma, \hat{P}_t \mid s)\rvert \le \Delta_t \coloneqq \tfrac{1}{2} R_{\max} H^2 \Delta_t^{L^1}$ for all $P \in \mathcal{P}_t$, where $\Delta_t^{L^1}$ is the largest L^1 radius of $\mathcal{P}_t$
2 NE transfer Lem. 3, Prop. 4	NE of $\mathcal{G}^\star$ with margin $\mu \Rightarrow (4\Delta_t - \mu)$-NE under every $P \in \mathcal{P}_t$. So found ⇒ ε-NE of $\mathcal{G}^\star$; not found ⇒ no NE has margin ≥ $4\Delta_t - \varepsilon$.
3 Coverage Lem. 6, 19, 20; Cor. 21	$p_t^e(P^\star) \ge p_{\text{reach}}$ + Freedman's inequality ⇒ w.p. ≥ 1 − δ/2 every (s,a) gets at least $n_{\min}$ visits

Main result

Theorem 7. W.p. ≥ 1 − δ, Alg. 1 terminates after at most

$n_{\min}$ visits per slot

$$\tilde{\mathcal{O}}\left(\frac{R_{\max}^2 H^4 |S|^2 |A|}{\varepsilon^2\, p_{\text{reach}}}\right)$$

Slots

Trajectories per episode

trajectory samples, and its output satisfies Def. 4, where

$$p_{\text{reach}} \coloneqq \min_{(s,a):\, s \notin S_T} \max_\sigma \inf_{P:\, \text{Supp}(P) = \text{Supp}(P^\star)} \mathbb{P}^{\sigma,P}[\text{reach } (s,a) \text{ before } S_T]$$

Infinite horizon: $\tilde{\mathcal{O}}\left(R_{\max}^2 |S|^6 |A| / (\varepsilon^2 p_T^4\, p_{\text{reach}})\right)$, where $p_T > 0$ is the minimum worst-case probability of reaching S_T within $|S|$ steps from any state.

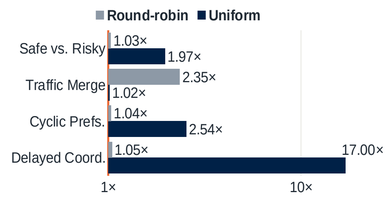
Experiments

PRISM-games · 6 benchmark CSGs · δ = 0.05

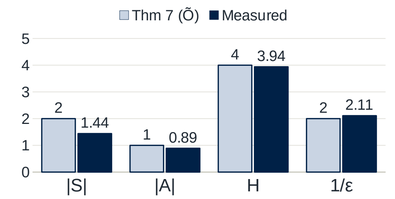
RQ1 · Correct outputs at termination (ε = 0.1)

	returns (a) ε-NE $\hat{\sigma}$	returns (b) NOT_FOUND
exact NE exists ($\bar{\mu} \ge 0$)	✓ **oracle's NE value attained** all 6 cases	— **never** (w.p. ≥ 1 − δ)
no exact NE	✓ **valid ε-NE** e.g. *Hide-or-run* ⇒ incomplete	✓ **sound certificate** e.g. *Cyclic Prefs.*

RQ2 · Exploration: samples vs. ours (1×)



RQ3 · Scaling (Safe vs. Risky variants)



Optimistic: ≈ same samples, but bigger value gap $V^\star - \hat{V}$ on Delayed Coord.

Takeaways & limitations

- Robust equilibria computed on the **learned model transfer** to the true game.
- NE **non-existence can be certified soundly** from data.
- **Limitations:** 1) assumes centralised exploration, known transition support, reachability; 2) scalability.

Paper Code

References [1] P. Bouyer, N. Markey, D. Stan. Mixed Nash equilibria in concurrent terminal-reward games. FSTTCS 2014.
[2] A. Y. He, D. Parker. Robust verification of concurrent stochastic games. TACAS 2026.

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arXiv:2609.04189 · github.com/7angel4/prism-games/tree/csg-learning